

100 Portfolio-Ready FDE Projects

Here are **100 portfolio-ready FDE projects**, progressing from foundational implementations to enterprise-grade AI deployments.

1. Customer Problem Solving & Rapid Prototyping

1. **Customer Support Ticket Triage System**

Automatically classify, prioritize, and route incoming support tickets.

2. **Intelligent FAQ Assistant**

Build a customer-facing assistant grounded in company documentation.

3. **Sales Lead Qualification Engine**

Score inbound leads and recommend which ones sales teams should prioritize.

4. **Employee IT Help Desk Assistant**

Answer common IT questions and escalate complex requests.

5. **Customer Feedback Analyzer**

Analyze surveys and reviews to identify themes, complaints, and opportunities.

6. **Meeting Action Item Tracker**

Extract decisions, owners, deadlines, and follow-up actions from meeting notes.

7. **Smart Email Routing System**

Categorize enterprise emails and forward them to the appropriate team.

8. **Contract Clause Extraction Tool**

Extract important clauses, dates, obligations, and risks from contracts.

9. **Resume-to-Job Matching Platform**

Match candidates against job requirements and explain fit.

10. **Enterprise Search Prototype**

Search across documents using semantic search instead of keywords.

2. APIs & Backend Systems

11. **REST API for Customer Management**

Build CRUD APIs with authentication, validation, and logging.

12. **Multi-Tenant SaaS Backend**

Design an API supporting multiple enterprise customers securely.

13. API Gateway Prototype

Route, authenticate, throttle, and log traffic across multiple services.

14. Webhook Processing Service

Receive events from external platforms and trigger internal workflows.

15. Rate-Limited Public API

Implement API keys, quotas, rate limits, and usage tracking.

16. Async Job Processing Platform

Build long-running background processing using queues and workers.

17. File Processing API

Upload PDFs, CSVs, and documents and asynchronously process them.

18. Authentication and Authorization Service

Implement login, RBAC, tokens, and permission enforcement.

19. API Integration Health Monitor

Detect failures and latency issues across third-party integrations.

20. GraphQL Enterprise Data Gateway

Expose multiple backend systems through a unified GraphQL interface.

3. Data Engineering Projects

21. CSV-to-Database Data Pipeline

Validate, clean, transform, and load customer data.

22. Automated Data Quality Checker

Detect missing values, duplicates, schema changes, and anomalies.

23. Customer 360 Data Pipeline

Combine sales, support, billing, and product usage data.

24. Incremental ETL Pipeline

Load only data that has changed since the previous run.

25. Real-Time Event Processing Pipeline

Process streaming customer or application events.

26. Data Deduplication Engine

Identify duplicate customers across disconnected databases.

27. Schema Mapping Assistant

Map customer database schemas into your product's canonical format.

28. Data Migration Validation System

Verify migrated datasets and detect inconsistencies automatically.

29. Data Lineage Tracker

Visualize where enterprise data originated and how it was transformed.

30. Automated Data Profiling Platform

Generate summaries and quality reports for unfamiliar customer datasets.

4. Integration Engineering

31. CRM Integration with Salesforce-Style APIs

Synchronize leads, accounts, and opportunities.

32. Slack Workflow Integration

Trigger enterprise workflows directly from Slack commands.

33. Microsoft Teams Notification Integration

Push alerts, approvals, and AI-generated summaries to Teams.

34. Google Workspace Integration

Connect Gmail, Drive, Docs, and Calendar into a business workflow.

35. ERP-to-Analytics Integration

Extract ERP transactions and expose operational dashboards.

36. HRIS Employee Synchronization System

Synchronize employee records between HR and internal applications.

37. Bi-Directional CRM Sync

Maintain consistent customer records between two systems.

38. Payment Platform Integration

Process payments, refunds, subscriptions, and payment webhooks.

39. Enterprise SSO Integration

Add SAML/OIDC authentication to an application.

40. Universal Connector Framework

Create reusable adapters for integrating multiple SaaS products.

5. Generative AI Applications

41. Document Q&A RAG Application

Ask questions across company policies, manuals, or technical documentation.

42. AI Customer Support Copilot

Suggest responses while grounding answers in company knowledge.

43. AI Sales Copilot

Generate account summaries, talking points, and recommended next actions.

44. AI HR Policy Assistant

Answer employee HR questions from approved internal policies.

45. AI Legal Document Assistant

Summarize contracts and identify potentially risky clauses.

46. AI Financial Report Analyzer

Ask natural-language questions across financial reports.

47. AI Requirements Generator

Convert customer discovery notes into requirements and acceptance criteria.

48. AI Incident Response Assistant

Analyze logs and suggest investigation steps during incidents.

49. AI Knowledge Base Generator

Convert scattered enterprise documentation into structured knowledge articles.

50. AI Executive Briefing Generator

Turn large amounts of business data into concise leadership summaries.

6. Advanced RAG Projects

51. Hybrid Search RAG Platform

Combine semantic search with traditional keyword search.

52. Multi-Document RAG System

Search across PDFs, spreadsheets, databases, and knowledge bases.

53. Role-Based RAG System

Restrict retrieved information based on employee permissions.

54. Citation-Aware RAG Assistant

Require answers to provide exact source references.

55. RAG Evaluation Framework

Measure retrieval accuracy, groundedness, relevance, and hallucinations.

56. Query Rewriting RAG System

Automatically improve ambiguous questions before retrieval.

57. Multi-Stage Retrieval Pipeline

Retrieve, rerank, filter, and generate enterprise answers.

58. Knowledge Graph + RAG Application

Combine structured entity relationships with unstructured documents.

59. Customer-Specific RAG Deployment

Give each enterprise customer an isolated knowledge base.

60. Self-Correcting RAG System

Detect weak retrieval results and automatically retry with improved queries.

7. Agentic AI Projects

61. Research Agent

Break a research objective into tasks, gather information, and create a report.

62. Customer Support Agent

Answer questions, access customer data, and create support tickets.

63. SQL Analytics Agent

Convert business questions into safe SQL queries and explanations.

64. Sales Operations Agent

Research accounts, update CRM records, and draft follow-ups.

65. Incident Investigation Agent

Analyze logs, metrics, and alerts and recommend remediation.

66. Document Processing Agent

Extract, validate, classify, and store uploaded documents.

67. Procurement Agent

Compare vendor proposals against company requirements.

68. Employee Onboarding Agent

Coordinate onboarding tasks across HR, IT, and management systems.

69. AI Workflow Approval Agent

Perform automated work but require human approval before sensitive actions.

70. Multi-Agent Enterprise Assistant

Route tasks to specialized HR, finance, IT, sales, and legal agents.

8. Enterprise Workflow Automation

71. Invoice Processing Automation

Extract invoice fields, validate them, and route exceptions for approval.

72. Purchase Order Approval System

Automate approval workflows based on business rules.

73. Employee Access Provisioning Workflow

Automatically provision or revoke system access.

74. Customer Onboarding Automation

Coordinate CRM, billing, support, provisioning, and communications.

75. Compliance Evidence Collection System

Automatically collect evidence for security or compliance audits.

76. Vendor Risk Assessment Workflow

Score vendors using questionnaires, documentation, and policy rules.

77. Customer Escalation Workflow

Detect high-risk customer issues and escalate them automatically.

78. Expense Approval Automation

Validate employee expenses and route them through approval rules.

79. SLA Monitoring and Escalation Engine

Detect tickets approaching SLA violations and trigger escalation.

80. Employee Offboarding Workflow

Coordinate account shutdown, access removal, and asset recovery.

9. Deployment, Reliability & Observability

81. Production FastAPI Deployment

Deploy an enterprise API with authentication, logging, and monitoring.

82. AI Application Observability Dashboard

Monitor latency, model usage, failures, tokens, and costs.

83. Centralized Logging Platform

Aggregate logs from multiple services and make them searchable.

84. API Performance Monitoring System

Track response times, throughput, failures, and availability.

85. Application Health Dashboard

Build operational dashboards for critical services.

86. Automated Failure Recovery System

Retry failed workflows and route unrecoverable errors appropriately.

87. Feature Flag Management Service

Enable controlled rollout of new functionality to customers.

88. Blue-Green Deployment Demonstration

Release a new application version without downtime.

89. AI Cost Monitoring Platform

Track AI usage and cost by customer, team, workflow, and model.

90. Usage Analytics Platform

Measure how enterprise customers actually use deployed functionality.

10. Security, Governance & Enterprise Architecture

91. Role-Based Access Control Platform

Implement users, roles, permissions, resources, and audit logs.

92. AI Prompt Injection Defense Gateway

Detect and block malicious instructions sent to LLM applications.

93. PII Detection and Redaction Service

Identify and remove sensitive information before processing.

94. Enterprise Audit Logging System

Record security-relevant actions across applications.

95. Secrets Management Prototype

Securely manage credentials used across integrations.

96. AI Model Evaluation Dashboard

Compare models on accuracy, latency, cost, and safety.

97. Human-in-the-Loop AI Review Platform

Route uncertain AI decisions to human reviewers.

98. Multi-Model AI Gateway

Route requests across multiple models based on price, speed, and capability.

99. Enterprise AI Governance Platform

Track AI applications, models, owners, risks, evaluations, and approvals.

100. End-to-End Forward Deployed Engineer Capstone

Take a fictional enterprise customer from **discovery** → **requirements** → **architecture**

→ integration → AI implementation → deployment → observability → ROI measurement.